Kartik Ramachandran

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Technical Skills

Mechanical Design: SOLIDWORKS, CATIA, Autodesk Inventor, Fusion 360, AutoCAD, OnShape
Manufacturing & Analysis: CNC Cutting, 3D Printing, Milling, Lathe, Welding, Root Cause Analysis, GD&T, FEA
Software: C++, Python, MATLAB, VEX PROS, VERILOG, Embedded Controls

Experience

Formula Electric — **September 2025 – Current**
Chassis Engineer — *University of Waterloo*

- Designing waterproof electronic enclosures for tractive system components to pass the FSAE rain test (2 minutes of direct spray with the tractive system live), targeting IP65 sealing through gasketed lids, sealed cable grommets
- Selecting enclosure materials and seals rated for vibration, heat, and moisture exposure per FSAE EV rules, ensuring rigid non-conductive covers that prevent contact with tractive system voltage during inspection and servicing
- Installed the brake pedal assembly, verifying rigid mounting to the chassis to meet the FSAE pedal force requirement
- 3D modelled and printed a joint tolerance measuring tool to verify weld angles at steel joints on the main chassis

PowerHouse Data Centre Group — **January 2026 – April 2026**
Data Centre Design Intern — *Calgary, Alberta*

- Developed site expansion drawings in AutoCAD for a TELUS data centre, integrating thermal, power, and computing infrastructure across redundant rooms to optimize floor space, cooling efficiency and power usage
- Engineered and specified thermal cooling solutions across 5 kW - 265 kW loads, supporting high-performance needs
- Produced electrical single-line diagrams for redundant power systems, tracing conditioned utility power through surge protection and UPS systems (rated 300 VA to 5,000 VA) to emergency backup batteries
- Conducted thermal unit configuration and testing for 20 kW to 500 kW facilities, comparing dry-bulb temperatures against static pressure to validate airflow, cooling performance and performance at required altitudes
- Created engraved lamacoid identification from electrical drawings for power cabinets at a High River, AB satellite facility, ensuring Canadian Electrical Code compliance for emergency identification

Projects

Exia Labs | *SOLIDWORKS, Fusion 360, Dynamics, Materials, Welding*

- Engineered a UGV for the NATO xTech Military Competition, carrying 200+ kg payloads across unstructured terrain using ROS-based path planning and 2D LiDAR obstacle detection in C++, winning $25K in competition funding
- Reverse engineered a 2007 Suzuki KingQuad ATV, performing a complete teardown to isolate and repurpose the drivetrain, transmission, throttle, and braking interfaces for drive-by-wire autonomous operation
- Designed and integrated an autonomous steering gearbox using ODrive motors, CNC'd steel gears, and a 3D printed housing, replacing the manual steering column to enable closed-loop front-wheel control
- Performed shear force and bending moment analysis to determine chassis support locations, then welded and angle-ground a steel frame validated to hold 200 kg at any point on the payload platform

210Z VEX Robotics | *Autodesk Inventor, Fusion 360, C++, CNC Cutting*

- Led end-to-end design, build, and integration of 25+ competition robots over three seasons, ranking first nationally in Canada, seventh in autonomous skills at the US Open, and finishing in the top 0.26% internationally
- Designed 6-motor drivetrains with PTO transmissions to dynamically reroute power between drive with intakes and catapult mechanisms, balancing RPM vs torque tradeoffs across 480 and 600 RPM gear ratios
- Developed and tuned closed-loop motion control in C++, implementing Pure Pursuit path tracking and 3-wheel odometry with inertial sensor fusion to improve autonomous pathing consistency
- Engineered a 3600 RPM silicone flywheel with a pneumatically adjustable launch ramp, modelling torque output vs release angle to achieve consistent scoring arcs

Education

University of Waterloo — **Expected April 2030**
Bachelor of Applied Science, Systems Design Engineering — *Waterloo, Ontario*